GENAI WORKS OPEN AGENT HACKATHON 2026 · WILDCARD [TINKERER]

QuizForge Live

The study agent that remembers what you missed.

Paste any study material — it plans a round, generates the questions, grades your answers, and adapts what comes next.

David Scott Cain · solo · Team HackOS

Built on QuizForge (v1, Build With AI: Basics, submitted Oct 23, 2026) — all Live work is new in this event window.

Demo video: 2:58 (film + browser Live-tab update) · Repo: github.com/dcain2336/quizforge-live

1 · PROBLEM + TARGET USERS

Quizzes forget the learner

- **The problem:** re-reading notes feels like studying but builds weak recall. One-shot quiz tools generate once, grade once, and adapt *never* — the learner's history evaporates between sessions.
- **The scaling trap:** an agent that re-reads the full history every round pays $O(n^2)$ context — planning gets slower and dumber as the learner improves.
- **Target users:** students and certification candidates studying from *their own* texts — our filmed demo uses a first-responder exam excerpt (scene safety, SAMPLE, OPQRST, DCAP-BTLS).
- **Also:** tutors and self-learners who need quizzes grounded in their material, plus an inspectable record of what the agent decided and why.

2 · SOLUTION

A study loop, not a quiz generator

- **QuizForge Live** turns any pasted study material into a live agent loop: **PLAN → GENERATE → GRADE → ADAPT**, round after round.
- Each plan is decided from a **bounded learner state** — mastered topics, remaining (weak) topics, strategy, difficulty — carried by the Meterless Markovian engine, never the raw history.
- Every miss returns an **evidence-backed explanation** that cites the study source; missed topics resurface next round until mastered.
- Every decision is **inspectable**: JSONL evidence (plan traces, grade telemetry, metered model calls) plus `/api/usage` and `/api/status` endpoints.
- The full loop runs **keyless and offline**; model routes enrich it when configured — it never blocks on credentials.

Plan → generate → grade → adapt

PLAN

agent.py + markovian.py fold graded history into the canonical state block (COMPLETED / REMAINING / DECISIONS / CONTEXT), compress it to an 800-token budget, and decide focus + difficulty.

→

GENERATE

The v1 QuizForge engine builds the round's questions from the material (LLM chain, then an offline engine that always works).

→

GRADE

Answers graded on the spot; explain.py attaches an evidence-backed explanation, citing the study sources, to every miss.

→

ADAPT

Missed topics land in the state's REMAINING list as telemetry — the next plan targets exactly them. Loop repeats.

When a model route is configured, Nemotron refines the plan — prompted with the bounded state via the engine's `build_prompt`, its reply marker-parsed and shape-validated, with the engine plan as the fallback. Metering (meterless.py) records every call and event as JSONL either way.

Zero dependencies, stdlib only

- **Runtime:** Python 3 standard library only — no pip installs, no build step; `python3 app.py` serves the UI and API on port 8765.
- **Modules:** `app.py` (HTTP API) · `agent.py` (the loop) · `markovian.py` (Meterless engine port) · `generator.py` (v1 engine) · `nemotron.py` (model routes) · `explain.py` (cited explanations) · `zetaris.py` (data-harness client) · `meterless.py` (JSONL evidence).
- **Models:** NVIDIA Nemotron (`nvidia/nemotron-3.5-super-vl:free`) via Nightshift locally or OpenRouter's free tier.
- **Data layer:** Zetaris REST — Bearer + X-Org-ID + X-Request-ID, SQL via `POST /sql-editor/sqls/run`.
- **Tests:** 30 deterministic offline tests (engine bounds, compression cascade, behavior change, env routing, full loop, and HTTP-level UI tests) — `python3 -m unittest discover -s tests`.

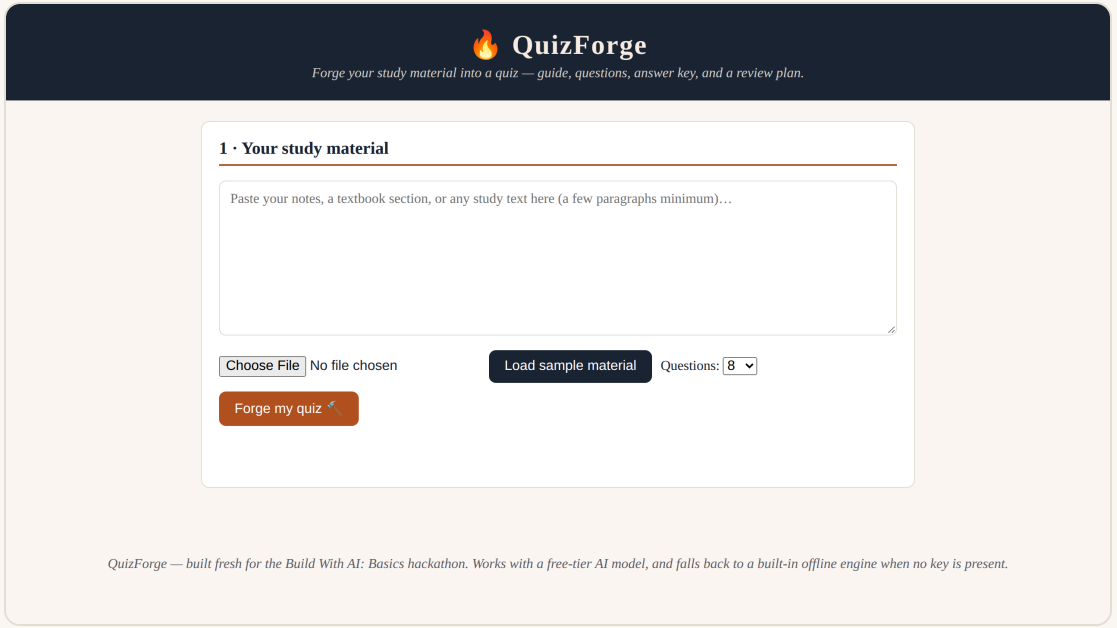
Sponsor technology, honestly labeled

- **Meterless — integrated and load-bearing.** The Markovian engine is ported from the open-source specs (Apache-2.0) and decides every plan; the Nightshift route is implemented (OpenAI-compatible `/v1/chat/completions`) and env-gated via `METERLESS_BASE_URL` / `METERLESS_RUNTIME_TOKEN` — Nightshift is local-install software with no hosted endpoint, so it activates on a participant machine.
- **Zetaris — client built, preflight-gated.** The full REST layer is implemented and verified against Zetaris's own public getting-started scripts, with their `SELECT 1` preflight pattern. The organizers' participant instance is still a TBC placeholder (no URL or key issued), so until one exists the corpus runs on the offline local-corpus fallback — and `/api/status` says exactly that.
- **Cursor — not used.** The organizers confirmed any coding tool is allowed and included no free Cursor credits in this event; we state this rather than claim it.

We didn't call the engine — we became it

- **A faithful port, not a wrapper:** upstream ships the Markovian engine as an implementation spec + TypeScript reference ("not a runtime library"), so `markovian.py` re-implements its core — config bounds, the `[STATE_CHECKPOINT]` marker protocol, the 4-level carryover cascade — in stdlib Python, with provenance and an honest adaptation list in the file header.
- **Learner state as a Markov chain:** topics transition weak → mastered by explicit rules. In the filmed session the carried state was **69 tokens**; the measured saving at 3 full chunks is 48,000 → 26,400 tokens (45%), growing with history — and the logs report 0 saved when a history is too small to save anything.
- **Evidence as a feature:** judges can replay every state transition from `logs/events.jsonl`. The demo's round-2 focus change is provable from the log, not just narrated over video.
- **Honesty as design:** `/api/status` exposes which sponsor pieces are live, gated, or fallback — no vaporware claims anywhere in the product.

The product, running offline



The app: paste any study material (or load the bundled first-responder sample). One command runs it — no installs, no keys required.


```
      "answer": "DCAP-BTLS",
      "topic": "DCAP-BTLS",
      "explanation": "Full statement: DCAP-BTLS stands for Deformities, Contusions, Abrasions, Punctu
    },
    {
      "type": "short",
      "question": "Fill in the blank: \"_____, Breathing & Circulation The airway priority order star
no trauma, or jaw thrust when trauma is suspected.\"",
      "options": [],
      "answer_index": -1,
      "answer": "Airway",
      "topic": "Airway",
      "explanation": "Full statement: Airway, Breathing & Circulation The airway priority order start
no trauma, or jaw thrust when trauma is suspected."
    }
  ],
  "plan": {
    "focus_areas": [
      "First Aid"
    ],
    "difficulty": "easy",
    "strategy": "First round; baseline probe of the session topic. Difficulty easy.",
    "engine": "markovian",
    "state": "COMPLETED: none yet\nREMAINING: none\nDECISIONS: First round; baseline probe of the ses
Aid. Attempts: 0 (0 correct, 0 missed). Engine: Meterless Markovian, carryover budget 800 tokens.",
    "state_level": "marker",
    "carryover_tokens": 61
  }
}
```


Round 1 (live API response): the Markovian plan — focus "First Aid", easy baseline probe, state COMPLETED: none yet — with the round's questions above it.

The adaptation, and its proof


```
      "answer": "SAMPLE",
      "topic": "SAMPLE",
      "explanation": "Full statement: SAMPLE stands for Signs and symptoms, Allergies, Medications, P
to the incident."
    },
    {
      "type": "short",
      "question": "Fill in the blank: \"_____ is used during a rapid trauma exam.\"",
      "options": [],
      "answer_index": -1,
      "answer": "DCAP-BTLS",
      "topic": "DCAP-BTLS",
      "explanation": "Full statement: DCAP-BTLS is used during a rapid trauma exam."
    }
  ],
  "plan": {
    "focus_areas": [
      "DCAP-BTLS",
      "SAMPLE"
    ],
    "difficulty": "easy",
    "strategy": "Revisit weak topics first: DCAP-BTLS, SAMPLE. Difficulty easy per the Markov state r
    "engine": "markovian",
    "state": "COMPLETED: Airway\nREMAINING: DCAP-BTLS, SAMPLE\nDECISIONS: Revisit weak topics first:
rule.\nCONTEXT: Session topic: First Aid. Attempts: 4 (1 correct, 3 missed). Engine: Meterless Markov
    "state_level": "marker",
    "carryover_tokens": 69
  }
}
```


Round 2's plan targets exactly the missed topics — focus: DCAP-BTLS, SAMPLE — decided from the state block: COMPLETED: Airway · REMAINING: DCAP-BTLS, SAMPLE (69 carryover tokens).


```
QUIZFORGE LIVE - EVIDENCE LOG (logs/events.jsonl, written by this session)
------------------------------------------------------------------------------------------
{"ts": "2026-10-23T18:20:24.023300Z", "session": "63e3bea2-602c-48a2-9523-c115432e28a4", "event": "tr
"plan_round", "topic": "First Aid", "attempts": 0, "focus_areas": ["First Aid"], "difficulty": "easy"
"history_tokens_estimated": 1, "tokens_saved_estimated": 0, "upstream": "github.com/Meterless/Meterle
markovian.py"}}
{"ts": "2026-10-23T18:20:24.029886Z", "session": "63e3bea2-602c-48a2-9523-c115432e28a4", "event": "te
"round_graded", "topic": "First Aid", "score": 1, "total": 4, "missed_topics": ["DCAP-BTLS", "SAMPLE"
{"ts": "2026-10-23T18:20:24.032382Z", "session": "63e3bea2-602c-48a2-9523-c115432e28a4", "event": "tr
"plan_round", "topic": "First Aid", "attempts": 4, "focus_areas": ["DCAP-BTLS", "SAMPLE"], "difficult
69, "history_tokens_estimated": 41, "tokens_saved_estimated": 0, "upstream": "github.com/Meterless/Me
markovian.py"}}
```


The session's evidence log (logs/events.jsonl): a trace per plan, telemetry for the graded round (score 1/4, missed topics recorded) — every state transition replayable.

Where Live goes next

- **Live Zetaris corpus** the moment a participant instance is issued: ingest a second source mid-session and watch the next round draw on it, with per-source provenance in explanations.
- **Nightshift on the participant's machine** (or a DGX Spark): local Nemotron planning through the already-built route; model checkpoints already parsed and kept as evidence.
- **More Meterless engines:** H-MEM for cross-session learner memory; Scout Intent to route across a multi-subject study library.
- **Spaced repetition from the state:** schedule reviews straight off the REMAINING list (v1's review plan meets the Markovian state), plus cohort/class mode and Anki export.
- **Hosted deployment:** a public demo instance, so anyone can play a round without cloning the repo — the zero-dependency app deploys as-is — plus persistent sessions, so a learner's Markovian state survives across visits.

QUIZFORGE LIVE

Plan · Generate · Grade · Adapt

Run it: `python3 app.py` → http://localhost:8765 — the whole loop works with no keys at all.

Repo: github.com/dcain2336/quizforge-live · Tests: 30 offline, all passing · License: MIT

GenAI Works Open Agent Hackathon 2026 · Wildcard [Tinkerer] · David Scott Cain (solo, team HackOS)

Pre-existing base: QuizForge v1 (declared). New in this event: the agent loop, the Markovian engine port, the Zetaris REST layer, the Nightshift route, and the JSONL evidence trail.